

Backend Quiz

JCWD-3302

1. What is the role of npm in Node.js application development?
 - a. Creating the layout of a website.
 - b. Handling HTTP request
 - c. Managing dependencies and packages
 - d. To store data in a database
2. How do you create a GET route endpoint in Express.js following proper REST conventions? (ex: for fetching a collection of products).
 - a. `app.get('/get-product', callback)`
 - b. `app.get('/products', callback)`
 - c. `app.post('/products', callback)`
 - d. `app.get('/product', callback)`
3. How do you configure middleware in Express.js? Assuming `app` is the variable declared for express.
 - a. `middlewareFunction.configure(app)`
 - b. `app.listen(middlewareFunction)`
 - c. `app.configure(middlewareFunction)`
 - d. `app.use(middlewareFunction)`
4. What is middleware in Express.js?
 - a. To handle authentication and authorization
 - b. To act as a filter between requests and responses
 - c. To process incoming requests before they reach the route handlers
 - d. All of the above
5. What is the difference between PUT and POST in the context of REST API?
 - a. PUT and POST can both be used to update data.
 - b. PUT and POST can both be used to create new data
 - c. PUT is used to update data, while POST is used to create new data.
 - d. PUT is used to create new data, while POST is used to update data.
6. What is the purpose of the select statement in Prisma queries?
 - a. Automatically join tables
 - b. Limit which fields are returned
 - c. Improve write speed
 - d. Cache the query result
7. What is the main purpose of an aggregate function?
 - a. To perform calculations on multiple rows
 - b. To join multiple tables together
 - c. To create new tables
 - d. To add indexes to rows
8. Which SQL function is used to count the total number of rows in a table?
 - a. `COUNT()`
 - b. `MAX()`
 - c. `SUM()`
 - d. `TOTAL()`
9. A REST API supporting pagination typically uses which two query parameters?
 - a. offset and limit
 - b. user and password
 - c. page and token
 - d. uuid and timestamp

10. What is the correct transaction block format in PostgreSQL?
 - a. TRY ... CATCH
 - b. BEGIN; ... COMMIT;
 - c. BEGIN; ... SAVE;
 - d. START; ... RETURN;
11. What is meant by a Foreign Key (FK) in SQL and how do you use it?
 - a. Foreign Key is used to identify unique records in a table.
 - b. Foreign Key is used to link two tables together based on a specific column.
 - c. Foreign Key is used to provide default values for certain columns.
 - d. Foreign Key is used to sort data in a table based on a specific column.
12. What is the correct SQL query to fetch all articles written by users with role "admin"?
 - a. SELECT * FROM articles WHERE role = 'admin';
 - b. SELECT a.* FROM articles a JOIN users u ON a.user_id = u.id WHERE u.role = 'admin';
 - c. SELECT users.* FROM users;
 - d. SELECT a., u. FROM users u;
13. How can you optimize the performance of a Node.js application handling high HTTP traffic?
 - a. Use serverless computing techniques to manage high traffic loads and ensure application availability
 - b. Avoid using clustering as it can cause unnecessary overhead and focus on optimizing the source code
 - c. Increase the use of synchronous programming to ensure data consistency, and use monitoring tools to detect bottlenecks
 - d. Use caching for frequently used query results and consider using a reverse proxy
14. Explain the difference between authentication and authorization in the context of creating REST API
 - a. Authentication is the process of granting access rights to a user, while authorization is the process of verifying the user's identity
 - b. Authentication is the process of verifying a user identity, while authorization is the process of granting access rights based on that verified identity.
 - c. Authentication and authorization are terms that can be used interchangeable because they have similar meanings.
 - d. Authentication and authorization are concepts used to store data in cache on the server.
15. How to create a new table with the condition of two columns where the first column is used to store numbers and the second column is used to store a name in PostgreSQL?
 - a. CREATE new_table (column1 INT, column2 VARCHAR(255));
 - b. CREATE new_table ADD COLUMN (column1 VARCHAR(255), column2 INT);
 - c. CREATE TABLE new_table (column1 VARCHAR(255), column2 INT);
 - d. CREATE TABLE new_table (column1 INT, column2 VARCHAR(255));
16. How do you delete all data from a table in PostgreSQL without deleting the table structure itself?
 - a. TRUNCATE table_name;
 - b. DROP table_name
 - c. REMOVE FROM table_name;
 - d. CLEAR * FROM table_name;
17. Explain the concept of normalization in the database design and why it is important?
 - a. Normalization is not important in database design because it can increase the complexity of SQL queries.
 - b. Normalization is the process of converting data into a simpler form. It is important to facilitate data readability in the database.
 - c. Normalization is the process of grouping data in a database based on its data type. It is important to ensure efficient data storage.
 - d. Normalization is the process of reducing redundancy and data dependencies in a database. It is important to ensure consistent data and reduce the risk of anomalies.
18. What is a nested subquery and how can you use it in SQL?
 - a. A nested subquery is a type of join operation that merges results from two tables.
 - b. A nested subquery is a query within another query, used to provide intermediate results for the outer query.
 - c. A nested subquery is a database trigger that automatically executes based on certain conditions.
 - d. A nested subquery is a type of stored procedure that can be called from within a function.

19. In Express.js, which error-handling pattern is correct?
- throw "Error message";
 - return res.error(message);
 - app.use((err, req, res, next) => { ... });
 - return next(req);
20. What is the purpose of the Prisma Migrate tool?
- To seed the database with random data
 - To test database queries
 - To generate and apply database schema changes based on the Prisma schema
 - To update environment variables

21. The API returns incorrect pages. Why?

```
await prisma.article.findMany({
  skip: req.query.page,
  take: 10
});
```

- take must be larger
 - skip should be (page - 1) * limit
 - findMany does not support pagination
 - Prisma requires cursor pagination
22. Which Express.js middleware is required to parse JSON request body?
- express.json()
 - express.static()
 - express.request()
 - express.file()

23. The query always returns null. Why?

```
const user = await prisma.user.findUnique({
  where: { name: req.params.name }
});
```

- Missing include
 - findUnique only works on unique fields
 - Use findMany only
 - You need to run a transaction
24. Which of the following data types is not supported by Prisma in the schema.prisma file?
- String
 - Datetime
 - Float64
 - Enum

25. Consider the TypeScript code above that uses Express JS for routing and middleware. What does the loggerMiddleware function do in this Express JS application?

```
index.ts

import express, { Request, Response, NextFunction } from 'express';

const app = express();
const port = 3000;

// Middleware
const loggerMiddleware = (req: Request, res: Response, next: NextFunction) => {
  console.log(`${new Date().toISOString()} ${req.method} ${req.url}`);
  next();
};

app.use(loggerMiddleware);

// Routes
app.get('/', (req: Request, res: Response) => {
  res.send('Hello, Express.js in TypeScript!');
});

app.get('/api/user/:id', (req: Request, res: Response) => {
  const userId = req.params.id;
  res.json({ userId, message: 'User details fetched successfully' });
});

app.listen(port, () => {
  console.log(`Server is running on http://localhost:${port}`);
});
```

- a. logs the timestamp, HTTP method, and URL of every incoming request to a file.
 - b. logs the timestamp, HTTP method, and URL of every outgoing response to the console.
 - c. logs the timestamp, HTTP method, and URL of every incoming request to the console.
 - d. handles authentication for protected routes using middleware.
26. If we successfully add a new record to our database, which HTTP response code is most appropriate to use? If we do not have proper access to access a request, what response code should be provided?
- a. 200 & 401
 - b. 201 & 500
 - c. 201 & 401
 - d. 200 & 500
27. What will the following Prisma query return?

```
const users = await prisma.user.findMany({
  where: {
    email: {
      endsWith: "@example.com",
    },
  },
});
```

- a. All users whose email ends with "@example.com"
- b. All users with verified emails
- c. All users with empty emails
- d. Syntax error due to invalid filter

28. Based on the three tables, select the answer below to get the names of students who have scores above the average for all courses taken?

```
CREATE TABLE Students (  
  student_id INT PRIMARY KEY,  
  student_name VARCHAR(255) NOT NULL,  
  date_of_birth DATE NOT NULL  
);  
  
CREATE TABLE Courses (  
  course_id INT PRIMARY KEY,  
  course_name VARCHAR(255) NOT NULL  
);  
  
CREATE TABLE Enrollments (  
  enrollment_id INT PRIMARY KEY,  
  student_id INT,  
  course_id INT,  
  grade DECIMAL(3, 2),  
  FOREIGN KEY (student_id) REFERENCES Students(student_id),  
  FOREIGN KEY (course_id) REFERENCES Courses(course_id)  
);
```

- SELECT s.student_name FROM Students s INNER JOIN Enrollments e ON s.student_id = e.student_id INNER JOIN Courses c ON e.course_id = c.course_id WHERE e.grade > (SELECT AVG(grade) FROM Enrollments GROUP BY student_id);
- SELECT s.student_name FROM Students s INNER JOIN Enrollments e ON s.student_id = e.student_id INNER JOIN Courses c ON e.course_id = c.course_id WHERE e.grade > (SELECT AVG(grade) FROM Enrollments GROUP BY course_id);
- SELECT s.student_name FROM Students s INNER JOIN Enrollments e ON s.student_id = e.student_id INNER JOIN Courses c ON e.course_id = c.course_id GROUP BY s.student_name WHERE AVG(e.grade) > (SELECT AVG(grade) FROM Enrollments);
- SELECT DISTINCT s.student_name FROM Students s INNER JOIN Enrollments e ON s.student_id = e.student_id INNER JOIN Courses c ON e.course_id = c.course_id WHERE e.grade > (SELECT AVG(grade) FROM Enrollments);